

Notes: Jung (RFS, 2025)

Real Consequences of Shocks to Intermediaries Supplying Corporate Hedging Instruments

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1 Big picture

- **Question.** Do shocks to financial intermediaries that supply corporate hedging instruments have real effects on nonfinancial firms?
- **Setting.** The paper studies South Korea around the 2010 macroprudential regulation that imposed limits on banks' foreign-exchange-derivatives (FXD) positions relative to capital. The evidence combines bank-level balance-sheet data, hand-collected firm-bank FXD contract data, and firm-level data on hedging and exports.
- **Core challenge.** A regulation that tightens bank constraints may affect many institutions at once, while the same macro shocks that motivated the regulation can also affect firms' demand for hedging. Even if bank FXD supply falls, firms may substitute to other banks, offshore dealers, futures, foreign-currency borrowing, or operational hedges. The paper therefore has to separate supply from demand and then show real effects at the firm level.
- **Main conceptual contribution.** The paper shifts attention from the standard credit-supply channel to a *hedging-supply channel*. Banks do not only allocate capital through loans. They also allocate risk through derivatives. If regulation or intermediary frictions reduce derivatives supply, firms can lose access to insurance and may alter real activity.
- **Main mechanism.** Exporters in Korea typically had long USD receivables from exports and hedged by selling USD forwards to banks. Banks became long USD forwards and financed the offsetting position with short-term foreign-currency borrowing. When regulation forced banks with high FXD-to-capital ratios to shrink positions, they reduced FXD supply rather than raising capital. Exporters that relied on those banks lost hedging capacity and optimally cut exports to reduce unhedged exchange-rate risk.
- **Main empirical design.** The paper traces the shock in three steps. First, it compares constrained and unconstrained banks in a difference-in-differences design. Second, it uses firm-bank contract data to compare hedging with constrained versus unconstrained banks for similar firms. Third, it examines whether firms exposed to constrained banks cut total hedging and then cut exports.
- **Main bank-level result.** Constrained banks reduced FXD positions substantially more than unconstrained banks after the regulation, while capital did not rise significantly. This is the key evidence that the policy worked mainly by shrinking derivatives books rather than by inducing new equity issuance.
- **Main contract-level result.** Exporters' hedging with constrained banks declined 47% more than their hedging with unconstrained banks, even after controlling for detailed contract, bank, and firm characteristics plus industry-location-size fixed effects.
- **Main firm-level hedging result.** Exposed exporters reduced total hedging by about 40%–45% relative to nonexposed exporters, with the effect concentrated in small firms. This means firms did not simply switch counterparties and fully offset the supply shock.
- **Main real result.** High-hedging exporters cut exports significantly more when they were more exposed to constrained banks. The preferred interaction estimate implies that a one-standard-deviation increase in exposure reduces export sales by about 16.5% more for high-hedge firms

than for low-hedge firms.

- **Key cross-sectional result.** Small firms are hit hardest on total hedging, and high-cash firms with high pre-shock hedge ratios cut exports by more, not less. That pattern is consistent with strategic risk management when hedging becomes more expensive, not with simple failure-to-hedge or speculation stories.
- **Main credibility results.** The paper argues against speculation, against a global credit-supply explanation, against general bank distress, and against a purely local demand story. Domestic sales do not fall, borrowing quantities and borrowing costs do not fall significantly, and long-tenor CIP deviations move in the direction implied by tighter FXD supply.
- **Economic takeaway.** Macroprudential FX regulation can transmit to the real economy through the supply of hedging instruments. The paper therefore establishes a direct causal link from intermediary constraints to corporate risk management and from corporate hedging to exports.

2 Data and measurement (key objects)

2.1 Economic context: why Korea is a useful setting

From the mid-2000s to the global financial crisis, Korean banks sharply increased short-term foreign-currency borrowing. Even after accounting for Korea's large stock of FX reserves, FX liquidity declined from about 18% of GDP to about 5% between 2004 and 2009. The banking sector then experienced a violent reversal in cross-border borrowing during the global financial crisis, making the policy problem concrete.

The institutional mechanism behind this vulnerability was closely tied to corporate hedging demand:

- Exporters earned USD receivables and hedged by *selling* USD forwards to banks.
- Banks therefore became *long* USD forwards.
- Because importer hedging demand was far smaller than exporter hedging demand, banks could not offset these positions naturally inside the corporate sector.
- Instead, they covered the positions through short synthetic forward positions funded by foreign-currency borrowing.

The paper reports that, in the first three quarters of 2007, exporters sold about USD 24 billion in forwards to banks, amounting to roughly 65% of banks' aggregate net long USD forward position. About 73% of the USD needed to fund those positions came from short-term cross-border borrowing.

Interpretation. This is exactly the environment in which a regulation aimed at bank FXD positions can spill over to exporters' risk management. The policy does not hit some side activity; it directly targets the balance-sheet technology banks were using to intermediate exporters' hedging demand.

2.2 The regulation: FXD position limit

The core policy rule is

$$\frac{|\text{FXD position}|}{\text{Capital}} < \text{Regulatory cap.} \quad (1)$$

The regulation was announced in June 2010 and initially became effective in October 2010. The initial caps were:

- 250% for foreign banks,
- 50% for domestic banks.

The policy was then tightened in 2011 and 2012 and loosened in 2016 and 2020. The main empirical analysis does not use the final 2020 loosening.

The paper emphasizes two measurement details:

- the FXD position is the monthly average of the daily net delta-adjusted notional value of all FX forwards, swaps, and options;
- the capital base includes Tier 1 capital and qualifying Tier 2 capital, including some longer-term parent funding for foreign bank branches.

Interpretation. The regulation is effectively a leverage-style cap on derivatives positions. Since banks can satisfy it either by cutting FXD or by raising capital, observing how they adjust is the first empirical step.

2.3 Data sources and sample construction

The paper combines three data sets:

- **Bank data.** Monthly FXD positions and related bank balance-sheet information, hand-collected from bank financial statements and regulatory data.
- **FXD contract data.** Hand-collected disclosures on derivatives contracts of listed nonfinancial firms from Korean corporate filings.
- **Firm data.** Balance-sheet, income-statement, and export information merged from Korean firm databases and filings.

The main bank sample contains 46 banks that were operating as of December 2009:

- 29 foreign banks,
- 17 domestic banks.

The contract-level data contain 251 firm-bank contracts among 132 firms and 33 banks, where a “contract” is defined as a firm-bank pair after aggregating all same-year positions between the pair. The broader firm sample has 148 firms that used FXD in both 2009 and 2010, though only 132 fully disclose all counterparties. The firm-level export analysis is therefore necessarily smaller than the bank panel.

Interpretation. The data are unusually rich because they let the author observe the exact transmission chain: bank exposure to the regulation, contract-level hedging with those banks, and firm-level hedging plus exports.

2.4 How the key variables are defined

Bank-level variables.

- *Constrained_i*: indicator for whether bank *i* exceeded the FXD-capital threshold when the regulation was introduced.
- *Shock_i*: the fraction of bank *i*'s FXD position that had to be reduced when the regulation was imposed.
- *logFXD*: log absolute FXD position.
- *logCapital*: log capital.

Contract-level variables.

- Net FXD notional for each firm-bank pair, scaled by firm assets in the main regressions.
- Contract direction distinguishes exporters' hedges (firms short foreign currency) from nonexporters' hedges (firms long foreign currency).
- Contract type and currency pair shares are used as controls.

Firm-level variables.

- *Exposed_j*: indicator that firm *j*'s main FXD counterparty bank is constrained.
- *Exposure_j*: notional-weighted average shock of firm *j*'s counterparty banks.
- *HighHedge_j*: indicator that the firm sold FXD worth more than 10% of export sales.
- Outcomes include changes in total net FXD position, export sales, domestic sales, borrowing, interest expense, and gross profit margin.

Interpretation. The paper uses both a binary treatment definition and a continuous exposure measure. That is helpful because the binary design is easy to interpret while the continuous design sharpens identification.

2.5 Measurement of net FXD positions

The paper computes each firm-bank pair's net FXD position as the delta-adjusted notional sum of all contracts between that firm and bank. For forwards, futures, and swaps, delta is 1. For options, the paper uses a conservative delta of 0.5 rather than a full Black–Scholes calculation.

A positive net FXD position means the firm is long foreign currency; a negative net position means the firm is short foreign currency. Exporters tend to have negative net FXD positions because they sell foreign currency forward to hedge export receipts.

Interpretation. This measurement choice matters because the policy directly targets the banks' net FXD books. The author needs a contract-level measure that lines up with the policy object.

2.6 Descriptive facts

The summary statistics already point toward the mechanism.

Bank data.

- Constrained banks had much larger FXD books relative to capital and assets.
- They were more likely to be foreign banks, smaller on average, more leveraged, and to have lower loan-to-asset ratios.
- The aggregate shock was economically large: constrained banks needed to cut about USD 15 billion of FXD positions, or roughly 40% of the constrained banks' initial positions.

Firm and contract data.

- The average firm-level net FXD position is about -8% of assets and about -10% of sales, meaning firms are on average short foreign currency through derivatives.
- The average export share is about 47% and the average export hedge ratio is around 0.41.
- The correlation between export sales and net FXD position is strongly negative at about -0.95 , while the correlation between foreign-currency liabilities and net FXD is strongly positive at about 0.78.

Interpretation. These correlations are very important. They show that FXD sellers are doing what an exporter hedging model predicts and FXD buyers are doing what a foreign-currency-borrower hedging model predicts. The patterns do not look like broad speculation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper's conceptual mechanism is: firms hedge exchange-rate risk to stabilize internal funds and reduce expected external financing costs. If hedging becomes more costly because intermediary supply contracts, firms optimally reduce the underlying risky activity—exports—rather than fully bearing the added risk.

This logic yields four clear empirical predictions:

- banks more constrained by the regulation should cut FXD positions more;
- firms dealing with those banks should lose hedging at the contract level;
- total hedging should fall most for firms that cannot easily substitute away, especially small firms;
- exports should fall most for firms that relied heavily on FXD hedging before the regulation.

The model also implies a distinctive cross-sectional prediction for internal funds. When hedging becomes more costly, firms with more internal funds can behave more cautiously and may cut exports by more, not less, if they are using exports strategically as a risk-management margin.

3.2 Bank-level difference-in-differences: constrained versus unconstrained banks

The baseline bank specification is

$$Y_{it} = \beta_0 + \beta_1 \text{Constrained}_i \times \text{Regulation}_t + \beta_2 \text{Constrained}_i + \gamma_t + \varepsilon_{it}. \quad (2)$$

A version with bank fixed effects is

$$Y_{it} = \beta_0 + \beta_1 \text{Constrained}_i \times \text{Regulation}_t + \delta_i + \gamma_t + \varepsilon_{it}. \quad (3)$$

The outcome Y_{it} is either $\log(\text{FXD})$ or $\log(\text{Capital})$. The time-varying regressor Regulation_t is the minimum FXD capital requirement, equal to zero before the regulation and increasing when the cap tightens.

Interpretation.

- If the regulation works mainly through shrinking derivatives supply, $\beta_1 < 0$ when $Y_{it} = \log(\text{FXD})$.
- If banks respond by raising equity instead, there should also be a significant positive effect on $\log(\text{Capital})$.

3.3 Event-study style check of pretrends

To examine the timing of the effect, the paper estimates

$$Y_{it} = \beta_0 + \beta_1 \text{Constrained}_i + \beta_2 \text{Post}_t + \sum_t \lambda_t \text{Constrained}_i \times \gamma_t + \varepsilon_{it}. \quad (4)$$

Here $Y_{it} = \log(\text{FXD})$ and the λ_t coefficients trace the relative gap in FXD positions between constrained and unconstrained banks month by month.

Interpretation. This is the graphical parallel-trends check. If the DiD design is credible, the constrained and unconstrained groups should evolve similarly before the first regulatory announcement.

3.4 Continuous-shock bank specification

The paper sharpens identification by replacing the binary treatment with a continuous exposure measure:

$$Y_{it} = \beta_0 + \beta_1 \text{Shock}_i \times \text{Regulation}_t + \delta_i + \gamma_t + \varepsilon_{it}. \quad (5)$$

Interpretation. Instead of asking whether a bank was constrained or not, this equation asks whether banks that needed to cut a larger fraction of their derivatives books responded more strongly. That is a cleaner dose-response design.

3.5 Contract-level transmission from banks to firms

The main contract-level regression is

$$\Delta \text{FXD}_{ij} = \alpha + \beta \text{Constrained}_i + \text{FirmControls}_j + \text{BankControls}_i + \text{ContractControls}_{ij} + \varepsilon_{ij}. \quad (6)$$

The continuous version is

$$\Delta \text{FXD}_{ij} = \beta_0 + \beta_{\text{Shock}} \text{Shock}_i + \text{FirmControls}_j + \text{BankControls}_i + \text{ContractControls}_{ij} + \varepsilon_{ij}. \quad (7)$$

The dependent variable is the change in the firm's net FXD position with bank i , scaled by firm assets, from 2009 to 2010. The author estimates these regressions separately for:

- exporter contracts, where firms are short foreign currency,
- nonexporter contracts, where firms are long foreign currency.

The preferred specifications add industry-location-size fixed effects to compare similar firms that happen to use different banks.

Interpretation. This is where supply is separated from demand. If constrained banks cut supply, the same type of firm should hedge less with constrained banks than with unconstrained banks.

3.6 Firm-level total hedging regressions

To test whether firms offset the loss of a constrained counterparty, the paper estimates

$$\Delta Y_j = \beta_E \text{Exposed}_j + \text{FirmControls}_j + \varepsilon_j, \quad (8)$$

and the continuous version

$$\Delta Y_j = \beta_E \text{Exposure}_j + \text{FirmControls}_j + \varepsilon_j. \quad (9)$$

The outcome is primarily the change in total net FXD position scaled by assets.

Interpretation. Contract-level losses alone do not prove that firms lose total hedging. Equations (8) and (9) test whether they can switch banks or use other margins to offset the shock.

3.7 Export regressions: high-hedge versus low-hedge firms

The main export equation is

$$\Delta Y_j = \beta_E \text{Exposure}_j + \beta_h \text{HighHedge}_j + \beta_{Eh} \text{Exposure}_j \times \text{HighHedge}_j + \text{FirmControls}_j + \varepsilon_j. \quad (10)$$

Here ΔY_j is the change in log export sales. The interaction term is the main parameter of interest.

Interpretation. The idea is simple: if hedging mattered for real activity, then firms that relied more on FXD before the shock should cut exports more when their banks are more exposed to the regulation.

3.8 Internal funds and the triple interaction

To study the role of internal funds, the paper estimates

$$\Delta Y_j = \beta_E \text{Exposure}_j + \beta_h \text{HighHedge}_j + \beta_c \text{Cash}_j \quad (1)$$

$$+ \beta_{Eh} \text{Exposure}_j \times \text{HighHedge}_j + \beta_{Ec} \text{Exposure}_j \times \text{Cash}_j \quad (2)$$

$$+ \beta_{hc} \text{HighHedge}_j \times \text{Cash}_j + \beta_{Ehc} \text{Exposure}_j \times \text{HighHedge}_j \times \text{Cash}_j + \text{FirmControls}_j + \varepsilon_j. \quad (11)$$

Interpretation. The triple interaction asks whether exposure hurts high-hedge firms differently depending on their internal funds. The sign of this coefficient is informative about whether the main mechanism is strategic risk management or a simple inability to absorb realized losses.

3.9 FXD pricing and CIP deviation

To provide suggestive price evidence that the regulation reduced *supply* rather than merely lowering *demand*, the paper defines covered-interest-parity (CIP) deviation for maturity n as

$$x_{t,t+n} = y_{t,t+n}^{\$} - \left(y_{t,t+n}^{\text{KRW}} - \frac{1}{n} (f_{t,t+n} - s_t) \right). \quad (12)$$

If constrained banks lower forward prices to discourage exporter hedging and shrink long USD-forward positions, long-maturity CIP deviations should move more than short-maturity deviations.

Interpretation. This is not the core identification strategy, but it is useful suggestive evidence that the FXD wedge moved in the direction predicted by a supply contraction.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no standard IV; identification comes from regulatory exposure heterogeneity

The paper does not use an external instrument in the standard two-stage-least-squares sense. Instead, identification comes from a quasi-natural experiment:

- the regulation was imposed nationally,
- but banks differed sharply in how binding it was,
- because some banks entered 2010 with much larger FXD positions relative to capital.

This is why the pre-shock DPTC ratio is central. It determines which banks were constrained and by how much they had to adjust.

Interpretation. The source of causal variation is not that some banks are regulated and others are not. It is that all banks are hit by the same rule, but the rule binds very differently depending

on preexisting exposure.

4.2 Why the bank-level design is credible

The bank-level DiD compares constrained to unconstrained banks over time. The most important threats are that constrained banks were already on different trends or were fundamentally different institutions.

The paper addresses this in three ways:

- the event-study coefficients are close to zero before the regulation,
- the main result survives bank fixed effects,
- and it survives continuous-shock specifications that flexibly use the dose of exposure.

The author also runs separate analyses for foreign banks and domestic banks, showing that the main bank-level response is not just a foreign-bank business-model story.

4.3 How the contract-level analysis separates supply from demand

A reduction in contract-level hedging could reflect either a fall in bank supply or a fall in firm demand. The paper's key move is to compare hedging with constrained and unconstrained banks for *similar firms*.

The preferred specifications therefore control for:

- firm size,
- pre-shock hedging,
- foreign-currency liabilities,
- bank size, leverage, and loans-to-assets,
- contract type and currency pair,
- industry-location-size fixed effects.

Interpretation. Once the same type of firm is compared across counterparties that differed in regulatory exposure, a relative decline in hedging with constrained banks is most naturally interpreted as a supply effect.

4.4 Why the firm-level results matter for identification

Suppose constrained banks reduce hedging, but firms easily replace them with unconstrained banks or offshore dealers. Then there is no reason for total hedging or exports to fall much.

The firm-level results matter because they show exactly the opposite:

- exposed exporters lose total hedging;
- the effect is especially strong for small firms;

- nonexporters are barely affected.

The paper interprets this as evidence that switching counterparties in FXD markets is costly, especially when derivatives are bundled with lending relationships or when contract customization is important.

4.5 Why the results are unlikely to be driven by speculation

A natural concern is that firms' derivative positions were speculative rather than hedging-oriented. The paper argues against this in three ways:

- exporters' net FXD positions are short USD while export revenues are effectively long USD, so the positions offset rather than amplify exposure;
- the correlation between export sales and net FXD position is strongly negative, about -0.95 ;
- exchange-rate risk is quantitatively large: exporters' total sales variance is many times the variance of domestic sales, and profit sensitivity to the exchange rate can be large when export shares are high and margins are thin.

Interpretation. These patterns make a pure speculation story very hard to maintain.

4.6 Why the credit channel is not the main explanation

A competing interpretation is that constrained banks were simply weaker during the global financial crisis and reduced *all* forms of credit, with exports falling because of lower borrowing rather than lower hedging.

The paper tests this directly and finds:

- no significant decline in firms' borrowing quantities,
- no significant increase in borrowing costs,
- no significant decline in banks' foreign-currency lending shares,
- and no effect of the regulation on domestic sales.

The author also shows that the export results remain when export changes are orthogonalized with respect to borrowing changes.

Interpretation. The evidence points much more strongly to a hedging-supply channel than to a general credit-supply channel.

4.7 Reputation risk and the decision to cut exports

The paper asks whether it is realistic for firms to reduce exports when export relationships may involve reputation concerns. The answer is yes for two reasons:

- firms can downsize future export contracts rather than renege on existing ones;
- the most affected firms are small manufacturers rather than global consumer brands, so reputa-

tional costs may be limited relative to the value of risk reduction.

Interpretation. Reputation risk does not overturn the main identification logic. If anything, it strengthens the message that hedging must be important if firms still cut exports.

4.8 Further robustness checks

The paper also addresses nonrandom firm-bank matching and sample imbalance.

- Exposed and nonexposed firms do not differ significantly on most observables.
- Correlations between exposure and key firm characteristics are small.
- Coarsened exact matching on the main potentially imbalanced characteristics yields similar estimates.
- The main firm-level sample is admittedly small, so the paper is careful to report this limitation explicitly.

Interpretation. The credibility of the paper comes from many mutually reinforcing pieces: exposure heterogeneity, contract-level supply tests, firm-level non-substitution, placebo tests, and robustness to matching-based sample reweighting.

4.9 Relation to the literature

The paper connects four literatures:

- intermediary constraints in FX markets,
- bank shocks and real activity,
- corporate hedging and firm value,
- macroprudential policy.

Its most distinctive contribution is to show that a bank regulation affecting derivatives provision can have real effects even when the traditional lending channel is weak or absent.

5 Tables and figures

5.1 Figure 1 and Figures 2–4: the macro setting and the hedging chain

Read:

- Figure 1 shows that the USD-KRW exchange rate is highly volatile, with annualized volatility reaching nearly 120% in the Asian crisis and still very high during later stress episodes.
- Figure 2 shows a sharp rise in banks' cross-border borrowing before the global financial crisis and a dramatic reversal in late 2008, with outflows near USD 40 billion in the fourth quarter of 2008.

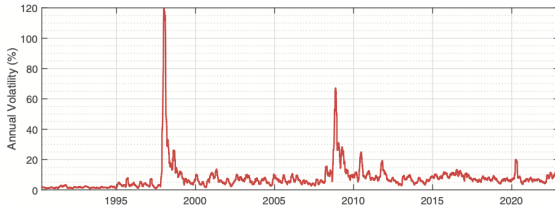


Figure 1
Volatility of USD-Korean won exchange rate
The annualized 30-day historical volatility of the USD-KRW exchange rate.

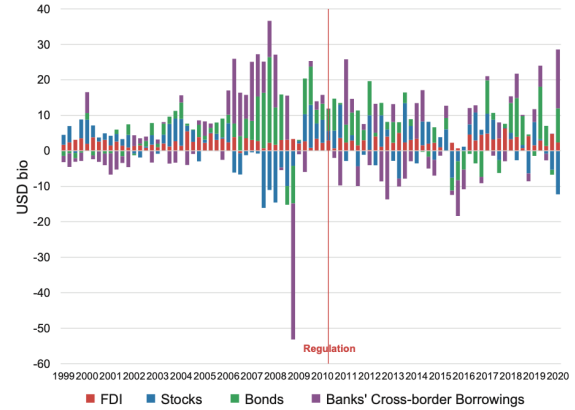


Figure 2
Gross foreign capital inflows
Korea's gross foreign capital inflows. The vertical line represents the imposition of the regulation.

- Figure 3 illustrates the balance-sheet chain: exporters are long foreign currency because of exports and hedge by selling FXD; banks are the counterparties and become long foreign currency through those FXD positions.
- Figure 4 shows that exporters accounted for about USD 24 billion, or 65%, of banks' net USD forward position, and that about 73% of the cover came from short-term cross-border borrowing.

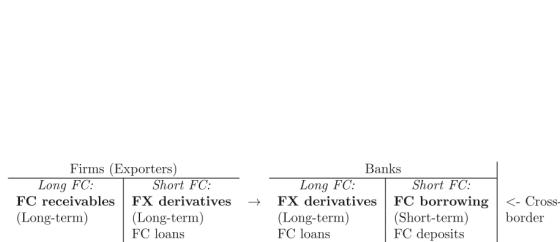


Figure 3
FX positions of exporters and banks before the regulation
The left panel illustrates the structure of the exporter FX position and the right panel illustrates the structure of the bank FX position prior to the regulation. Exporters had a long position in foreign currency (because of export sales) and hedged the exposure by taking short positions in FXD. As banks are firms' FXD counterparties, banks had a long position in foreign currency due to the FXD. Banks hedged the long exposure by foreign currency borrowing.

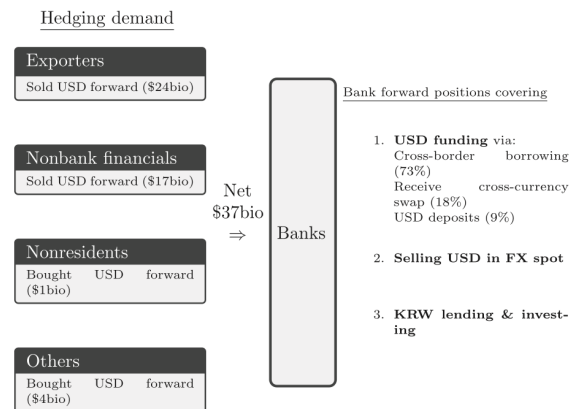


Figure 4
FX forward hedging demand and bank positions covering
The left panel shows the amount of USD forwards each nonbank sector purchased from or sold to banks during the first three quarters of 2007. The right panel shows how banks covered the net long position of US\$37 billion in forwards. Source: Bank of Korea (2008).

Economic message. These figures explain why the regulation matters. The banking system was using short-term external funding to intermediate corporate FX hedging on a large scale, making derivatives supply directly relevant for macro-financial vulnerability.

5.2 Table 1, Tables 2–4, and Figures 5–6: regulation and exposure in the data

Read:

- Table 1 lays out the time path of the FXD caps: foreign banks move from 250% to 200% to

Announced on	Jun. 13, 2010	May 19, 2011	Nov. 27, 2012	Jun. 16, 2016	Mar. 18, 2020
Effective from	Oct. 31, 2010	Jul. 31, 2011	Jan. 31, 2013	Jul. 31, 2016	Mar. 19, 2020
Foreign banks (%)	250	200	150	200	250
Domestic banks (%)	50	40	30	40	50

The top two rows show the announcement dates and effective dates. The regulation was first announced on June 13, 2010. The bottom two rows show the historical changes in the regulatory cap on the ratio of FXD to capital. A limit of 250% means that a bank's FXD position is required to be lower than 2.5 times its capital.

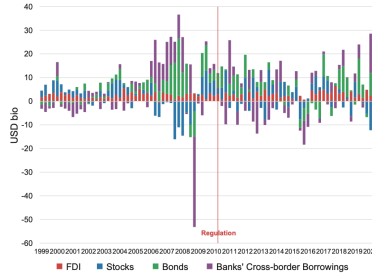


Figure 2
Gross foreign capital inflows
Korea's gross foreign capital inflows. The vertical line represents the imposition of the regulation.

Firms (Exporters)		Banks		
Long FC:	Short FC:	Long FC:	Short FC:	
FC receivables	FX derivatives	FX derivatives	FC borrowing	← Cross-border
(Long-term)	(Long-term)	(Long-term)	(Short-term)	
	FC loans	FC loans	FC deposits	

Figure 3
FX positions of exporters and banks before the regulation
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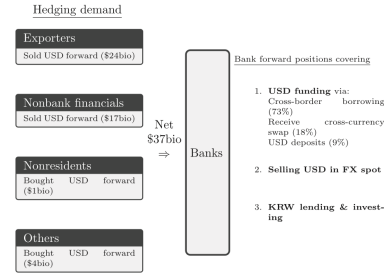


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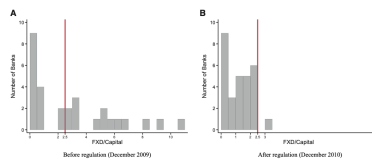


Figure 5
FXD position to capital ratio, before and after the regulation (foreign banks)
The histogram of the FXD position to the capital (DPTC) ratio of foreign banks, 6 months before and 6 months after the first announcement of regulation. The vertical line represents the regulatory cap on the DPTC ratio.

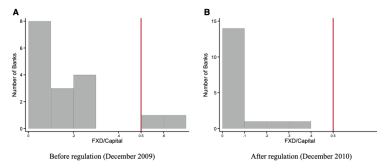


Figure 6
FXD position to capital ratio, before and after the regulation (domestic banks)
The histogram of the FXD position to the capital (DPTC) ratio of domestic banks, 6 months before and 6 months after the first announcement of regulation. The vertical line represents the regulatory cap on the DPTC ratio.

150%, while domestic banks move from 50% to 40% to 30%, before later partial loosening.

- Table 2 shows that constrained banks had average FXD/assets of 31% versus 5% for unconstrained banks, and were smaller and more leveraged.
- Table 3 shows that about 40% of firm-bank contracts were with constrained banks and that contract type and currency composition differ across constrained and unconstrained contracts.
- Table 4 shows that exposed and nonexposed firms look broadly similar on most observables, though the sample is not huge.
- Figures 5 and 6 show that many banks above the cap before the regulation moved below it afterward: 14 foreign banks and 2 domestic banks were initially above the cap and almost all adjusted beneath it.

Economic message. Before any regression is run, the raw data already show meaningful cross-bank heterogeneity in how binding the regulation was and meaningful exposure of firms to those banks.

5.3 Figure 7, Figure 8, and Table 5: bank-level response to the regulation

Read:

- Figure 7 shows that aggregate FXD positions fell sharply after the regulation and remained lower, while normalized FXD positions of constrained banks diverged downward relative to unconstrained banks as the policy tightened.
- Figure 8 is the event-study plot. The differential coefficients are near zero before the regulation and then become clearly negative afterward.

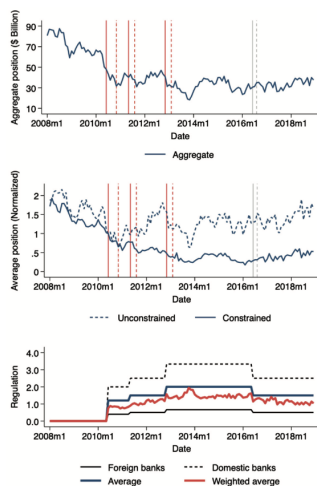


Figure 7
FXD position by treatment
The top panel plots the aggregate FXD position. The vertical solid (dotted) lines indicate the announcement (effective) dates of changes in the minimum FXD capital requirement, with red (gray) lines indicating tightening (loosening) adjustments. The middle panel plots the normalized average FXD position of constrained (solid) and unconstrained (dotted) banks. The average FXD positions are normalized such that they equal one at the imposition of the regulation. The bottom panel plots the minimum FXD capital requirements. The blue line represents the simple average, and the red line represents the weighted average minimum FXD capital requirements where the weight is FXD position.

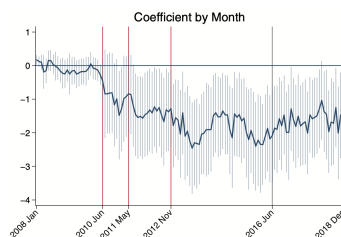


Figure 8
Coefficient by month
Plot of β_t over time in the following specification: $Y_t = \beta_0 + \beta_1 \text{Constrained}_t + \beta_2 \text{Pos}_t + \sum_{i=1}^4 \beta_{i+1} \text{Constrained}_t \times \gamma_i \gamma_{i+1}$, where Pos_t takes the value of one for the time period after the regulation (June 2010) and zero otherwise. γ_i is the time dummy variable for each month. The vertical lines represent the adjustments of the regulatory cap. The first three vertical lines (in red) represent tightening adjustments, and the last vertical line (in gray) represents a loosening adjustment.

Table 5
Impact on bank FXD position and capital

A. Based on simple average FXD capital requirement				
	(1)	(2)	(3)	(4)
	logFXD	logFXD	logCapital	logCapital
Constrained-1 x Regulation	-.912*** (-3.18)	-.967*** (-3.28)	.0288 (.33)	.0362 (.34)
Constrained-1	5.341*** (3.92)		-.646 (-1.52)	
Bank FE	N	Y	N	Y
Time FE	Y	Y	Y	Y
N	5,906	5,906	5,885	5,885
Adj. R-sq.	.109	.802	.055	.915
t-statistics are in parentheses. *p < .1, **p < .05, ***p < .01				
B. Based on weighted average FXD capital requirement				
	(1)	(2)	(3)	(4)
	logFXD	logFXD	logCapital	logCapital
Constrained-1 x Regulation	-1.207*** (-3.05)	-1.292*** (-3.17)	.0203 (.19)	.0152 (.15)
Constrained-1	5.312*** (3.91)		-.631 (-1.49)	
Bank FE	N	Y	N	Y
Time FE	Y	Y	Y	Y
N	5,906	5,906	5,885	5,885
Adj. R-sq.	.109	.803	.055	.915
t-statistics are in parentheses. *p < .1, **p < .05, ***p < .01				

The regressions in this table examine the impact of the regulatory shock on the bank FXD position. The top panel uses $\text{Regulation}^{\text{simple}}$, which takes zero before the regulation and the simple average of foreign bank and domestic bank minimum FXD capital requirements. The bottom panel uses $\text{Regulation}^{\text{weighted}}$, which is the weighted average of the minimum FXD capital requirements, where the weight is the FXD position in each month. In either case, a higher value indicates tighter constraint. Columns 2 and 4 add bank fixed effects. The sample period is 2008–2019 on a monthly basis. Standard errors are clustered by bank. All variables are defined in the appendix.

- Table 5 reports the core DiD estimates. Using the simple average regulation measure, the constrained-bank coefficient is about -0.913 to -0.967 for $\log(\text{FXD})$ and insignificant for $\log(\text{Capital})$.
- Using the weighted-average regulation measure, the coefficients are even larger in magnitude, about -1.207 to -1.292 for $\log(\text{FXD})$, again with no significant capital response.

Economic message. The regulation made constrained banks shrink derivatives supply rather than recapitalize. This is the foundational empirical fact of the paper.

5.4 Table 6 and the contract-level transmission results

Read:

- For exporter contracts, the coefficient on *Constrained* is 0.0529 without controls and 0.0374 with controls, implying a large reduction in hedging with constrained banks.
- The paper interprets the controlled exporter-contract estimate as about a 47% larger decline in hedging with constrained banks than with unconstrained banks.
- For nonexporter contracts, the coefficients are near zero or very small.
- The result remains after adding industry-location-size fixed effects and after excluding option contracts.

Economic message. The contract data are where the paper really nails down the supply channel. The response is concentrated where the regulatory constraint bites most: exporter-facing bank positions.

	(1) Exporters	(2) Exporters	(3) Nonexporters	(4) Nonexporters	(5) Full sample	(6) Full sample
Constrained	.0529*** (3.66)	.0374** (2.52)	.00189 (1.00)	.00317*** (2.09)	.0228** (2.28)	.0129* (1.70)
Type: Swaps		.0114 (.59)		-.00114 (-.15)		.00511 (1.13)
Type: Options		.0862*** (4.48)		0 (.)		.0992*** (6.38)
Type: Futures		.0111 (.54)		0 (.)		.00293 (.34)
Pair: EURKRW		.0661 (1.20)		0 (.)		.0469 (1.45)
Pair: JPYKRW		-.0188 (-1.29)		.00658** (2.17)		.00104 (.15)
Pair: XXXKRW		-.00541 (-.43)		-.00207 (-.18)		-.000744 (-.13)
Firm controls	N	Y	N	Y	N	Y
Bank controls	N	Y	N	Y	N	Y
N	129	129	122	122	251	251
R-sq.	.0964	.353	.00419	.125	.0371	.315

t-statistics are in parentheses.

p* < .1, *p* < .05, ****p* < .01

The regressions in this table examine the impact of the regulatory shock on firm FXD contracts. I define contract as a firm-bank pair. The dependent variable is the change in net FXD position dealt between firm *j* and bank *i* between 2009 and 2010, scaled by assets. *Constrained_{it}* takes the value of one if the contract is dealt with a constrained bank and zero otherwise. Firm controls include log size, net FXD notional (scaled by sales) before the shock, foreign currency liability share, and seven industry dummies. Bank controls include log size, loans-to-assets ratio, leverage ratio, and a foreign bank indicator variable. Contract controls include bank *i*'s share of firm *j*'s total FXD notional, type, and currency pair. The omitted categories are forwards and USD-KRW pair. Standard errors are clustered at the bank level. All variables are defined in the [appendix](#).

5.5 Tables 7–8 and Tables B.7–B.8: firm-level total hedging

Read:

- In Table 7, the coefficient on *Exposed* for the full sample is about 0.0385 with controls, while for small firms it rises to 0.0716 and for large firms it is only about 0.0091.
- In Table 8, exposed exporters have a coefficient of 0.0728 with controls, while nonexporters have coefficients very close to zero.
- In the fully disclosed subsample, Table B.7 shows that the continuous exposure coefficient is 0.0265 in the full sample and 0.0331 for small firms.
- Table B.8 shows that the same exposure coefficient is about 0.0507 for exporters and essentially zero for nonexporters.

Table 7
Impact on firm-level FXD position by firm size, full sample

	(1) Full sample	(2) Full sample	(3) Small	(4) Small	(5) Large	(6) Large
Exposed	.0352** (2.13)	.0385** (2.43)	.0608** (2.50)	.0716** (2.49)	.00838 (.40)	.00910 (.52)
Constant	-.00329 (-.28)	.0265 (.17)	-.00167 (-.10)	-.180 (-.24)	-.00487 (-.28)	-.260 (-.98)
Firm controls	N	Y	N	Y	N	Y
N	148	148	74	74	74	74
R-sq.	.0253	.0771	.0743	.186	.00151	.0237

t-statistics are in parentheses.

p* < .1, *p* < .05, ****p* < .01

The regressions in this table examine the impact of the regulation on firm-level FXD positions. The outcome variable is change in firm *j*'s net FXD position scaled by assets. Independent variable *Exposed* takes the value of one if the firm's main FXD counterparty bank is constrained. Firm controls include log size, net FXD notional (scaled by sales) before the shock, foreign currency liability share, and seven industry dummies. All variables are defined in the [appendix](#).

Table 8
Impact on firm-level FXD position by net FXD position, full sample

	(1) Exporter	(2) Exporter	(3) Nonexporter	(4) Nonexporter
Exposed	.0640** (2.48)	.0728*** (2.72)	-.00226 (-.39)	-.00229 (-.41)
Constant	-.00302 (-.17)	-.0811 (-.27)	-.00380 (-1.24)	.0451 (.84)
Firm controls	N	Y	N	Y
N	92	92	56	56
R-sq.	.051	.113	.00307	.0798

t-statistics are in parentheses.

p* < .1, *p* < .05, ****p* < .01

The regressions in this table compare the impact of the regulation on the firm-level FXD positions of exporters and nonexporters. A firm is classified as an exporter (nonexporter) if it holds a negative (positive) net FXD position. The outcome variable is the change in firm *j*'s net FXD position scaled by assets. The independent variable *Exposed* takes the value of one if the firm's main FXD counterparty bank is constrained. Firm controls include log size, net FXD notional (scaled by sales) before the shock, foreign currency liability share, and seven industry dummies. All variables are defined in the [appendix](#).

Economic message. These tables show that total hedging really falls at the firm level and that the effect is concentrated exactly where the substitution problem should be most severe: small exporters.

Table B.7

Impact on firm-level FXD position by firm size, fully disclosed firms

	(1) Full sample	(2) Full sample	(3) Small	(4) Small	(5) Large	(6) Large
Exposure	.0235*** (3.45)	.0265*** (3.86)	.0320*** (2.83)	.0331*** (3.13)	.0152** (2.27)	.0170*** (2.73)
Constant	.00976 (1.29)	.145 (1.26)	.0179 (1.37)	-.154 (-.23)	.00240 (.31)	.222 (1.11)
Firm controls	N	Y	N	Y	N	Y
N	132	132	66	66	66	66
R-sq.	.0687	.164	.0888	.465	.0537	.154

t-statistics are in parentheses.

* $p < .1$, ** $p < .05$, *** $p < .01$

The regressions in this table examine the impact of regulation on firm-level FXD positions when I restrict the sample to firms that fully disclosed their FXD counterparty information. The outcome variable is the change in firm j 's net FXD position scaled by assets. Independent variable *Exposure* is the weighted average shock of the firm's FXD counterparty banks. Firm controls include log size, net FXD notional (scaled by sales) before the shock, foreign-currency liability share, and seven industry dummies. All variables are defined in the [appendix](#).

Table B.8

Impact on firm-level FXD position by net FXD position, fully disclosed firms

	(1) Exporter	(2) Exporter	(3) Nonexporter	(4) Nonexporter
Exposure	.0447*** (3.95)	.0507*** (4.22)	.00131 (.45)	.000644 (.19)
Constant	.0231* (1.98)	.0166 (.07)	-.00569** (-2.08)	.0590 (.88)
Firm controls	N	Y	N	Y
N	82	82	50	50
R-sq.	.14	.245	.00502	.0851

t-statistics are in parentheses.

* $p < .1$, ** $p < .05$, *** $p < .01$

The regressions in this table compare the impact of regulation on the firm-level FXD positions of exporters and nonexporters when I restrict the sample to firms that fully disclosed their FXD counterparty information. The outcome variable is the change in firm j 's net FXD position scaled by assets. A firm is classified as an exporter (nonexporter) if it holds a negative (positive) net FXD position. Independent variable *Exposure* is the weighted average shock of the firm's FXD counterparty banks. Firm controls include log size, net FXD notional (scaled by sales) before the shock, foreign currency liability share, and seven industry dummies. All variables are defined in the [appendix](#).

5.6 Tables 9–12: export responses, placebo outcomes, and internal funds

Table 9

Impact on export sales

A. High hedge versus low hedge firms

	(1) logExport	(2) logExport
High hedge=1 $\times$ Exposure	-.199* (-1.94)	-.165* (-1.81)
Exposure	.0498 (.77)	.0834 (1.55)
High hedge=1	.143 (1.34)	.0271 (.30)
Constant	.210** (2.58)	-1.618 (-1.23)
Firm controls	N	Y
N	74	74
R-sq.	.0817	.324

t-statistics are in parentheses.

* $p < .1$, ** $p < .05$, *** $p < .01$

B. Continuous hedge ratio

	(1) logExport	(2) logExport
Exposure $\times$ Export hedge ratio	-.176*** (-3.96)	-.212** (-2.24)
Exposure	-.0499 (-1.02)	-.0479 (-.85)
Export hedge ratio	.0887 (1.35)	.164** (2.26)
Constant	.301*** (6.17)	-1.660 (-1.30)
Firm controls	N	Y
N	74	74
R-sq.	.228	.464

t-statistics are in parentheses.

* $p < .1$, ** $p < .05$, *** $p < .01$

The regressions in this table examine the impact of the regulation on exports. The outcome variable is the change in log export sales. The independent variable *Exposure_{it}* is the weighted average shock of firm j 's FXD counterparty banks. The top panel uses *HighHedge_{it}*, which takes the value of one if firm j sold amount of FXD is more than 10% of its export sales, and zero otherwise. The bottom panel uses *ExportHedgeRatio_{it}*, which is firm j 's sold amount of FXD divided by export sales. Firm controls include log size, net FXD notional (scaled by sales) before the shock, foreign currency liability share, and seven industry dummies. All variables are defined in the [appendix](#).

Table 10

Impact on firm-level FXD position

	(1) FXD/Asset	(2) FXD/Asset
High hedge=1 $\times$ Exposure	.0518*** (2.70)	.0581*** (2.84)
Exposure	.0108 (1.22)	.0108 (1.16)
High hedge=1	.0401** (2.04)	.0413** (2.06)
Constant	-.00855 (-1.06)	-.125 (-.56)
Firm controls	N	Y
N	74	74
R-sq.	.215	.319

t-statistics are in parentheses.

* $p < .1$, ** $p < .05$, *** $p < .01$

The regressions in this table examine the impact of the regulation on the firm-level FXD position. The outcome variable is the change in firm j 's net FXD notional scaled by assets. All variables are defined in the [appendix](#).

Read:

- In Table 9, the interaction *Exposure* $\times$ *HighHedge* is -0.199 without controls and -0.165 with controls. The paper interprets column (2) as implying that a one-standard-deviation increase in exposure lowers export sales by about 16.5% more for high-hedge firms than for low-hedge firms.
- The continuous-hedge-ratio version in Panel B yields a similarly strong interaction, around -0.212 with controls.
- Table 10 shows that high-hedge firms also experience a much larger reduction in total FXD positions when exposed to more constrained banks; the interaction is 0.0581 with controls.
- Table 11 is the placebo test using domestic sales. The interaction coefficient is small and insignificant, around -0.008 with controls.

- Table 12 adds cash. The triple interaction is -3.327 for export sales and -1.301 for export share, implying that high-cash, high-hedge firms cut exports more when more exposed.

Table 11
Impact on domestic sales as a placebo test

	(1) logDomesticSales	(2) logDomesticSales
High hedge=1 × Exposure	-.0324 (-.37)	-.00794 (-.09)
Exposure	-.00657 (-.09)	.000843 (.01)
High hedge=1	.128 (1.46)	.0935 (.95)
Constant	.0888 (1.25)	.315 (.35)
Firm controls	N	Y
N	74	74
R-sq.	.0353	.118

t-statistics are in parentheses.

* $p < .1$, ** $p < .05$, *** $p < .01$

The regressions in this table examine the impact of the regulation on domestic sales. The outcome variable is the change in firm j 's log domestic sales. All variables are defined in the [appendix](#).

Table 12
Role of internal funds on export sales and export share

	(1) logExport	(2) ExpShare
High hedge=1 × Exposure × Cash	-3.327** (-2.10)	-1.301*** (-3.72)
High hedge=1 × Exposure	.0817 (.68)	.0596** (2.26)
High hedge=1 × Cash	-1.677 (-1.29)	-.983*** (-3.35)
Exposure × Cash	2.196 (1.63)	.837** (2.56)
Exposure	-.0162 (-.22)	-.0199 (-.94)
High hedge=1	.0780 (.58)	.0403 (1.56)
Cash	2.844** (2.42)	1.030*** (3.66)
Constant	-2.588** (-2.22)	-.631** (-2.63)
Firm controls	Y	Y
N	74	74
R-sq.	.423	.527

t-statistics are in parentheses.

* $p < .1$, ** $p < .05$, *** $p < .01$

The regressions in this table examine the impact of the regulation, hedge ratio, and internal funds (cash balance) on export sales and share of export sales. The outcome variable is export sales or share of export sales. Independent variable $Exposure_j$ is the weighted average shock of firm j 's FXD counterparty banks. $Cash_j$ is firm's preshock cash and cash equivalent balance scaled by total assets. Firm controls include log size, net FXD notional (scaled by sales) before the shock, foreign currency liability share, and seven industry dummies. All variables are defined in the [appendix](#).

Economic message. The export results are not generic firm distress. They line up tightly with the hedging channel: high-hedge firms lose more hedging and cut exports, while domestic sales do not move in the same way.

5.7 Table 13, Figure A.4, Figure A.5, and the appendix tables on alternative explanations

Table 13
Sensitivity of the impact of exchange rates on profits

	Case	Export sales (USD value)	Profitability (USD value)	FC liability share (USD value)	Profit (USD value)
A	Base level (Export share 50%, profit margin 20%) After a 1% depreciation of USD	\$10,440 \$10,440	1,605 1,275	12,143 12,143	9,402 8,326
B	Base level (Export share 50%, profit margin 20%) After a 1% depreciation of USD	\$10,440 \$10,440	1,605 1,355	12,143 12,143	9,402 9,174
C	Base level (Export share 50%, profit margin 20%) After a 1% depreciation of USD	\$10,440 \$10,440	1,605 1,355	12,143 12,143	9,402 8,326

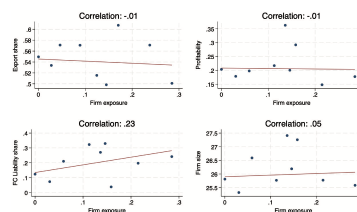


Figure A.4
Correlations between firm characteristics and firm exposure
Binned examples of firm characteristics (export share, profitability, FC liability share, and firm size) and firm exposure to the regulation.

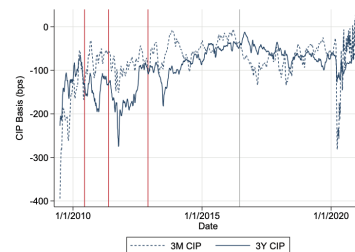


Figure A.5
CIP deviations: Short term and long term
Ten-day moving average of 3-year (solid) and 3-month (dotted) USD-KRW CIP deviations, where CIP deviation is defined in Equation (12). The first three red vertical lines represent tightening adjustments, and the last gray vertical line represents a loosening adjustment.

Read:

- Table 13 shows how sensitive profits can be to exchange rates. In the high-export-share example based on Samsung Heavy, a 1% USD depreciation lowers profit by about 10.7%.
- Figure A.4 shows low correlations between firm exposure and key firm characteristics such as export share, profitability, foreign-currency liability share, and firm size.
- Figure A.5 shows that long-term CIP deviations fell relative to short-term deviations after the first tightening announcements, consistent with tighter forward supply at long maturities.
- Appendix Tables on borrowing and foreign-currency lending show no significant decline in firms' borrowing or banks' foreign-currency loan shares.

Economic message. The appendix does three jobs: it shows the FX risk is economically important, it shows exposure is not obviously driven by firm characteristics, and it provides extra evidence against both the credit-channel story and the demand-only interpretation.

6 Short summary

- The paper studies how a bank regulation that restricts FX derivatives positions affects bank hedging supply, firm hedging, and firm exports.
- The Korean setting is ideal because exporters heavily used direct FX derivatives, and banks covered the resulting positions with short-term foreign-currency funding that the regulation targeted directly.
- Constrained banks cut FXD positions rather than raising capital, establishing that the policy tightened the supply of hedging instruments.
- Exporters dealing with constrained banks reduced hedging much more than similar exporters dealing with unconstrained banks, which identifies a bank-supply effect rather than a pure demand effect.
- Exposed exporters, especially small ones, were unable to fully substitute away and therefore reduced total hedging.
- Firms that relied heavily on FXD before the shock cut exports more when they were more exposed to constrained banks, while domestic sales did not respond in the same way.
- The evidence is inconsistent with a broad credit-supply story and difficult to reconcile with speculation. It points to a causal hedging channel from intermediary regulation to real trade outcomes.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that shocks to financial intermediaries supplying hedging instruments have real effects. In the South Korean quasi-natural experiment, leverage-style FXD regulation reduced banks' derivatives supply, and that reduction propagated to firms that relied on those banks for hedging. Those firms, especially small exporters with high hedge ratios, cut total hedging and then cut exports.

7.2 Economic intuition

The key intuition is that hedging is not just a side financial decision. For exporters facing large exchange-rate risk, hedging changes the volatility of internal funds and therefore the expected cost of future financing. When hedging becomes more costly because banks retreat from the market, firms do not simply accept more risk. They can also reduce the risky activity itself. In this paper, that activity is exports.

A second intuition is about what banks do in the economy. They do not only lend. They also stand between firms and risk. When regulation makes that intermediation more expensive, the real consequences can show up through the risk-management margin rather than the classic credit margin.

7.3 Takeaway

This paper's message is that intermediary frictions in derivatives markets can matter for trade just as intermediary frictions in credit markets matter for investment. More broadly, it shows that macroprudential regulation can have meaningful nonfinancial side effects when it constrains the private supply of risk-sharing instruments.